

Elyas Assili

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Applied Mathematics and Computer Science Engineer, pursuing a second degree in Quantitative Finance and Risk Management. I have gained diverse professional experiences in France and internationally, working with global leaders, research laboratories, and SMEs. I am seeking a permanent position in Quantitative Finance or Global Market for 2026.

EDUCATION

Panthéon Sorbonne <i>Master's degree in Risk Engineering: Finance and Insurance (M2) — Major: Financial Mathematics</i>	Sept 2025 – June 2026 Paris, France
Polytech Sorbonne – École Polytechnique Universitaire de Sorbonne Université <i>5-year Engineering Degree — Major: Applied Mathematics and Computer Science</i>	Sept 2021 – Nov 2024 Paris, France
École de Technologie Supérieure <i>Study Abroad Program — Major: Artificial Intelligence</i>	Sept 2023 – Dec 2023 Montréal, Canada

EXPERIENCE

HSBC – Asset-Liability Management Quantitative Analyst - Engineered portfolio analysis workflows processing diverse financial inputs (model points, credit ratings, outputs from economic scenario generators). - Conducted comprehensive ALM studies (AoC, duration, dilution, spread) to inform asset allocation decisions. - Architected and deployed the migration of a complex ALM projection model from Excel into a scalable Python calculation engine. - Manually rewrote advanced financial formulas to optimize the ORSA reporting pipeline via direct CSV ingestion, supporting Solvency II compliance. - Developed a custom GUI for rapid visualization of ALM datasets stored as .pickle files. - Built an advanced regex-based parser to recursively map cell dependencies across legacy Excel and VBA workbooks.	Sept 2025 – Present Paris, France
Thales – Data and Analytics Engineer - Developed a data decoding and analysis tool to process and interpret large-scale system information with Lua and C. - Designed an interactive mapping system for data visualization and anomaly tracking with Python.	Jan 2024 – July 2024 Paris, France
Kyoto University, Mathematics Research Laboratory – Quantitative Analyst / Quantitative Researcher - Built mathematical and computational tools for stochastic modeling and numerical optimization. - Applied quantitative finance methods (Monte Carlo simulations, stochastic processes, model calibration). - Conducted advanced statistical analysis and automated complex simulations using C++, Python and MATLAB. - Created an original system of mathematical equations for modeling (Black-Scholes, Heston, Hull-White, Volatility, VaR). - Delivered training sessions for multidisciplinary international experts.	June 2023 – Aug 2023 Kyoto, Japan
Snap Inc. (Snapchat) – Data Scientist - Designed and implemented advanced artificial intelligence and data science solutions to support strategic decision-making. - Developed, optimized, and rigorously backtested machine learning models for financial time-series forecasting, signal extraction, and state inference. - Built end-to-end data pipelines, including preprocessing, feature engineering, and model evaluation. - Applied advanced statistical methods and quantitative analysis to model volatility, assess risk, and evaluate alternative market datasets. - Utilized a wide range of AI and data science tools and frameworks within an international MLOps environment.	Sept 2022 – May 2023 Remote
Nomad advantage (Nomadia) – Data and Analytics Engineer Applied mathematical and statistical methods to analyze datasets, identify trends, and generate actionable insights.	June 2022 – Aug 2022 Paris, France

PERSONAL PROJECTS & RESEARCH

- Optimal Trade Execution (Quantitative Research Paper) — HJB stochastic control for optimal order liquidation under the Square-Root Law; +20% vs TWAP.
- Financial Market Risk Calculator & Dashboard — R/Shiny dashboard computing VaR & ES via Extreme Value Theory (POT/GARCH) with crash backtesting.
- Ultra-Fast Trading Platform & Real-Time Dashboard — Multi-threaded C# Limit Order Book with market-making bot and live Blazor/SignalR visualization.
- Quant AI Bot (RAG Chatbot for Quantitative Finance) — Local LLM chatbot (Ollama/LlamaIndex) ingesting research papers for context-aware quant finance queries.
- Investment Portfolio Optimizer & Dashboard — Markowitz MPT optimizer with efficient frontier visualization and dynamic backtesting (Python/Streamlit).
- Financial Options Pricing & Risk Calculator — C++ engine for Black-Scholes/Monte Carlo pricing and full Greeks computation (Delta, Gamma, Vega...).
- Automated Trading Bot for Correlated Assets — Automated statistical arbitrage bot based on cointegration analysis with full backtesting pipeline (Python).
- AI Tool for Extracting Receipt Totals — Computer vision + NLP model (PyTorch) for automatic total price extraction from scanned receipts.

LEADERSHIP & AWARDS

- JPMorgan Chase & Co. Quantitative Certificate
- Engaged in student associations: Enactus (social entrepreneurship), Polytech Without Borders (international solidarity), and SOPhIA (artificial intelligence).
- 1st Prize — Hackathon on Artificial Intelligence and Musical Improvisation.
- Selected among 1,400 high school students to participate in the ERASMUS+ program in Poland.

TECHNICAL SKILLS

Python, R, C, C++, C#, MATLAB, Java, VBA, SQL, SAS, ERP, GCP, PostgreSQL, Power BI, Artificial Intelligence tools, LaTeX, Office, Excel, PowerPoint, Git, Docker, Talend

SOFT SKILLS

Analytical Thinking · Team-player · Rigorous · Ability to Work Under Pressure

LANGUAGES

French: Native · English: Fluent · Arabic: Conversational · Spanish: Conversational · Japanese: Notions

INTERESTS

Cinema · Chess · Volleyball